MASPOE's

MVP Poll application

3th BSc Business Engineering

Database systems & Data Structures

OVERVIEW

01
About Maspoe

02
Why this project?

03
Bussines Idea

04
User journey

05
Core features

06
Demonstration

Our Team

Judith Ceuppens
General Manager

Louis Van Linden
Backend Engineer

Arjen Sap
Database Architect

Korneel Pennewaert
Frontend Developer

Kasper De Vos
Database Architect

ABOUT MASPOE

- Belgian event-organizing organization
- Focused on sports, culture and entertainment
- Organises lots of events/festivals
- Biggest event: Vicaris Village

MASPOE
SPORTS & EVENTS

Why this project?

Festival line-ups are typically decided with limited and unstructured feedback

Organizers lack a structured data model to capture preferences

Existing feedback channels (social media, comments) are noisy and non-quantifiable

No mechanism to translate user preferences into actionable insights

Business Idea

A poll-based platform that structures user input into quantifiable data

Visitors influence the line-up through controlled voting mechanisms

Organizers obtain aggregated and comparable results to support booking decisions

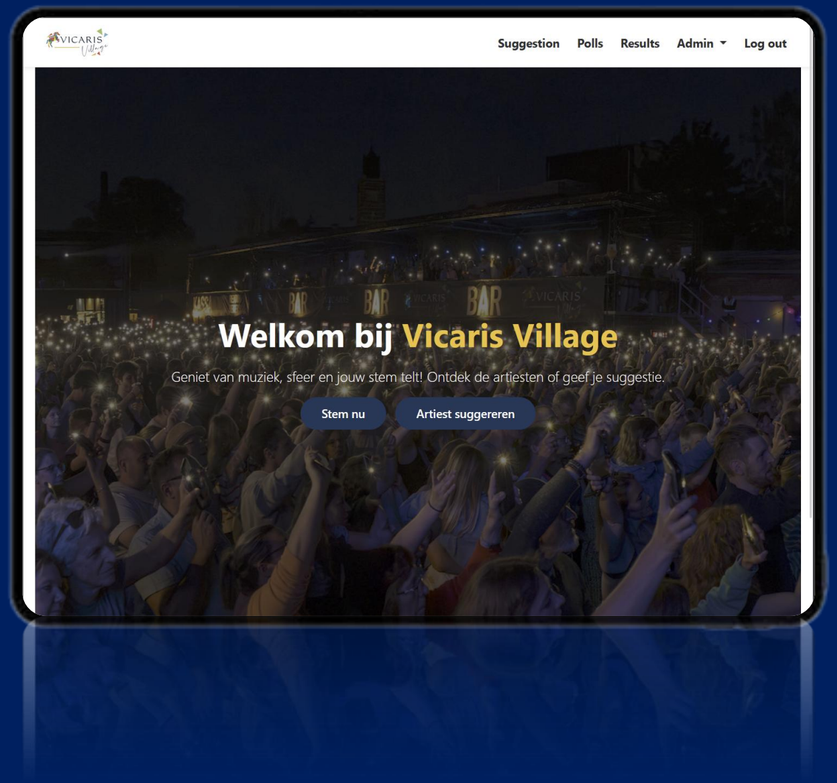
User journey- visitor

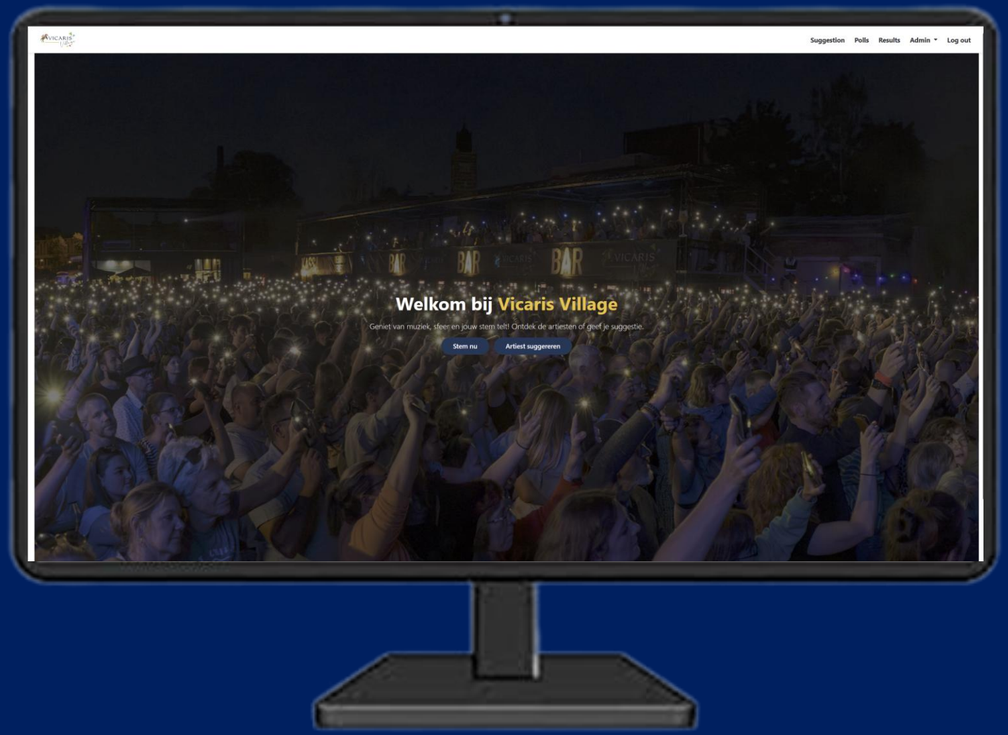
User journey- organizer

Core features

- The platform allows users to suggest and vote for artists in a structured manner
- Polls are generated based on predefined criteria and user input, ensuring relevance
- Voting results are automatically aggregated and stored consistently
- Organizers can access an overview of poll outcomes through the application

Compatible on tablet and desktop





Extra features

Administrators can manage artists and festival editions, including adding and removing entries

The application enforces role-based access, separating user and admin functionalities

Results are presented in a clear and interpretable format to support decision-making

The underlying data structure allows the platform to be reused for future festivals and editions

Demo

https://1drv.ms/v/c/2b685717d1176f31/IQBXTGSAPZziSo6jY0ZeQqT5ARf0T71pGdcN3ttObiWkPfU?e=5kPvPL

Thank you for listening

Group 38

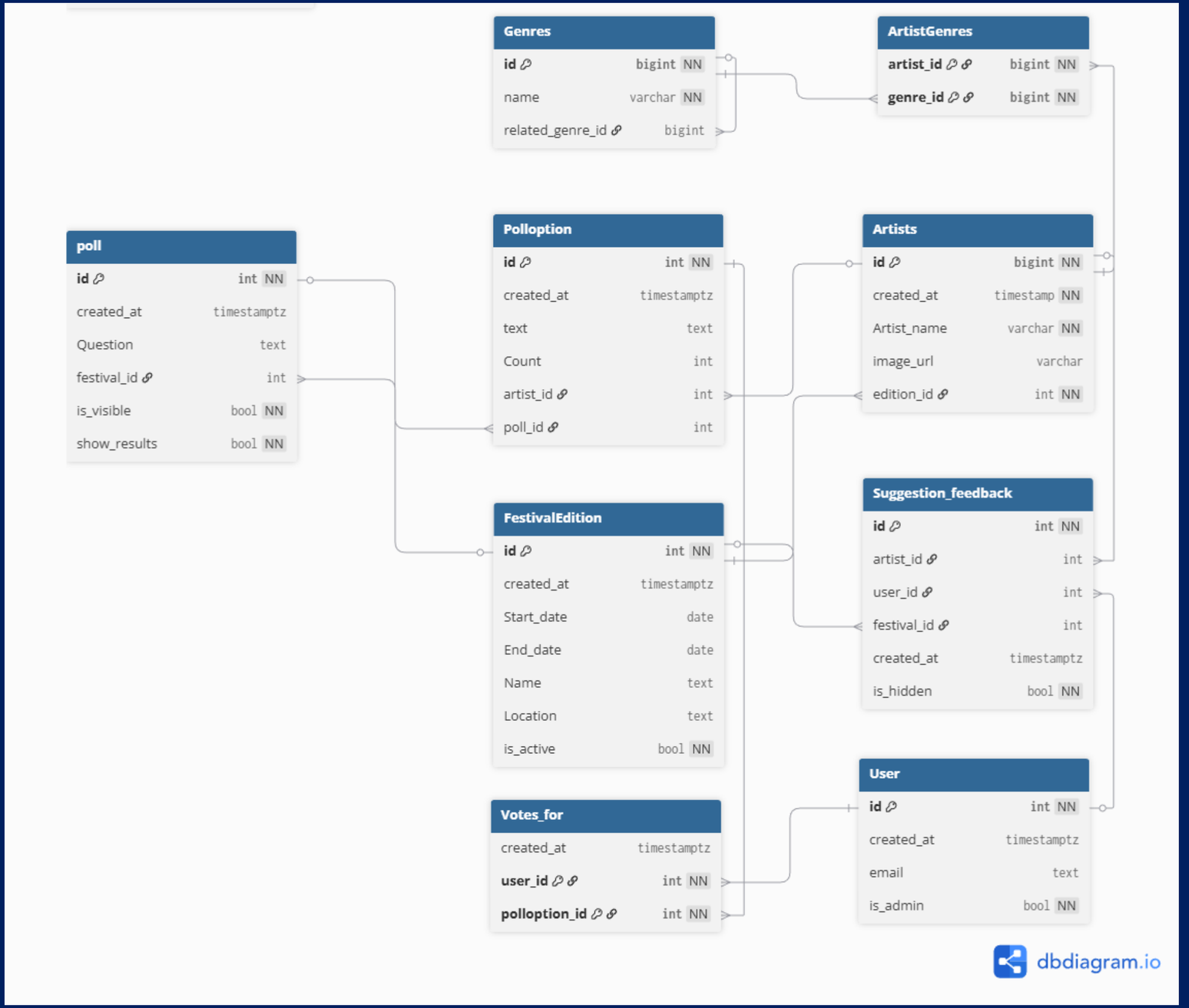
Q&A

User stories

- As a user, I want to view available polls and submit a vote or suggestion, so that I can actively participate in the Vicaris Village selection process.
- Users can submit a vote via the polling interface
- Users can suggest an artist using a form
- As a user, I want to view clear and structured poll results and submitted suggestions, so that I can understand the collective preferences of the community.
- As an admin, I want to manage artists, editions and polls, so that the Vicaris Village application remains accurate, up to date and scalable for future events.

ERD

- 9 entities
- 42 attributes
- Each entity has 1 or more primary keys
- Relationships implemented using foreign keys
- N:M relationships resolved using junction tables



Overall architecture

- Flask *application factory* (`create_app()`)
- Central `models.py` using SQLAlchemy
- Blueprints in `app/routes/`
- HTML templates in `app/templates/`
- Static assets in `app/static/`
- Business logic in `app/services/`
- Helper utilities in `app/utils/`

Flask app

- `create_app()` loads configuration (`Config`)
- Initializes database (`db.init_app`)
- Enables Flask-Migrate
- Registers all blueprints
- Context processors:
  - `current_user`
  - `poll_visibility`
  - `suggestions_visibility`
  - `results_visibility`
- @app.before_request-hook: always an active poll

Routes

- `core.py` --> homepage
- `auth.py` --> register / login / logout
- `suggestions.py` -->artist suggestions (max. 5 per edition)
- `poll.py` --> poll loading, voting, and results
- `admin.py` --> admin dashboard
- `admin_editions.py` --> edition management

Database models

- `User` --> authentication + `is_admin`
- `FestivalEdition` --> edition details + active flag
- `Artists` — artists linked to an edition
- `Genres` + `ArtistGenres` --> many-to-many relationship
- `SuggestionFeedback` --> user suggestions per edition
- `Poll`, `Polloption`, `VotesFor` --> complete poll system

Services and business logic

- `poll.py` service:
  - `get_active_festival()`
  - `get_or_create_poll_for_edition()`
  - `get_or_create_active_poll()`
- `genre_profile.py` service:
  - Builds user music profile from suggestions
  - Generates personalized poll options
- `session.py` utility:
  - `get_session_user()`

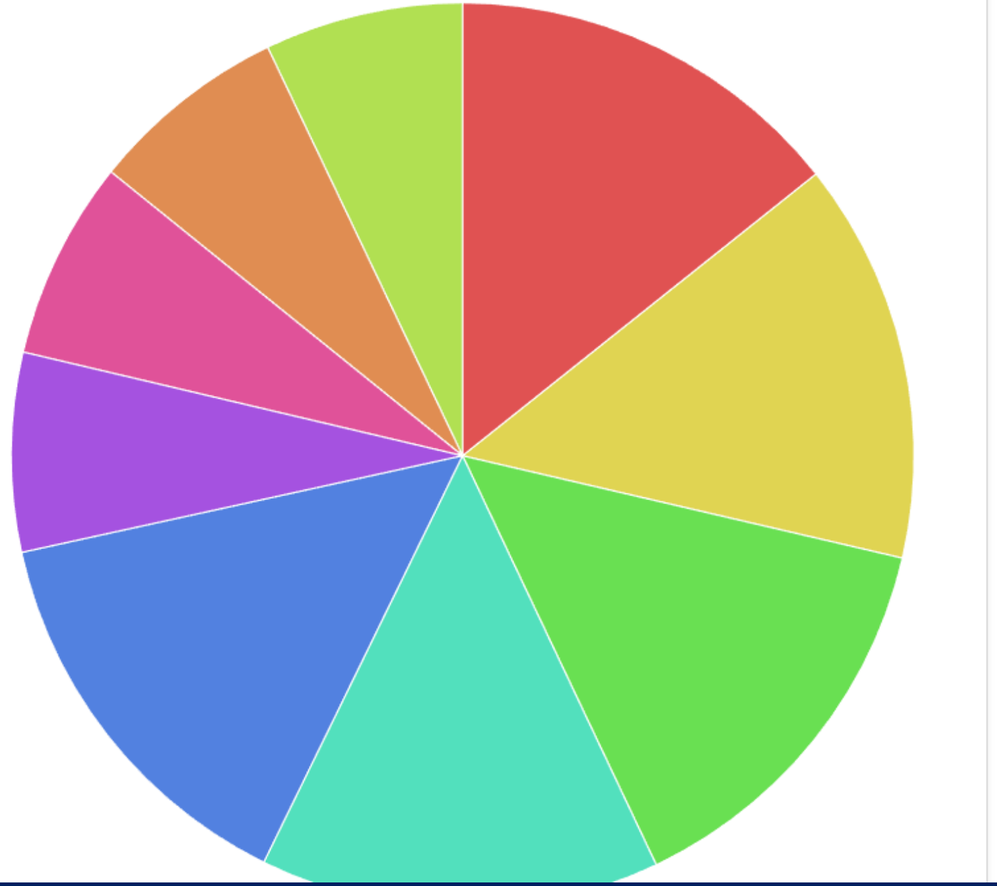
Algorithmic support

- Builds a user genre profile based on suggestions
- Scores artists using genre similarity
- Selects top-ranked artists for personalised polls

Strenghts

- Useful for festivals
- Easy to use for customers
- Easy to filter artists for admin
- Upscaleable
- Everyone can vote
- Sorted into genres

Result



Tekstueel overzicht (toegankelijkheidsfallback)

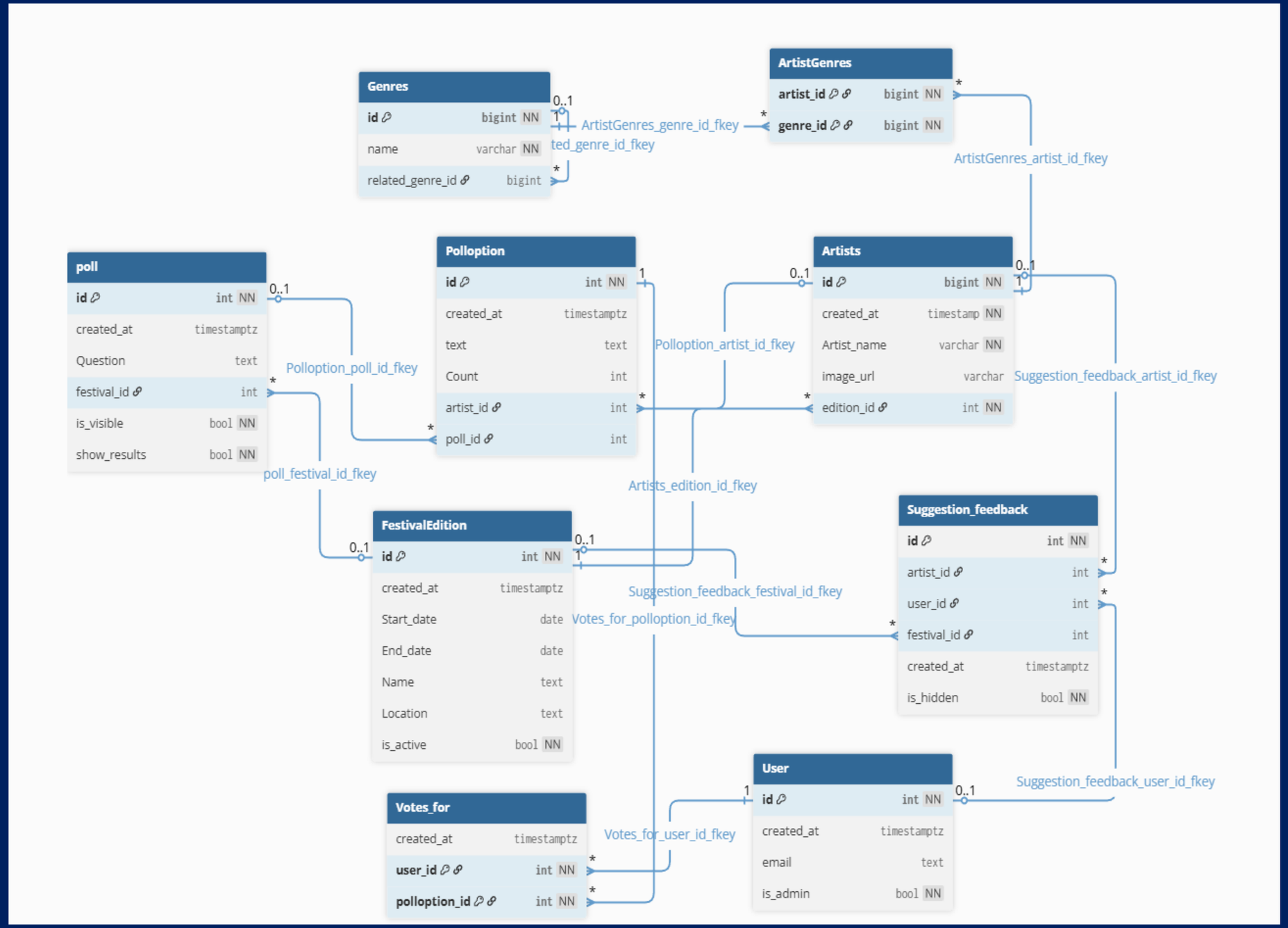
Hip-Hop	**14.3% (2)**
Pop	**14.3% (2)**
Rock	**14.3% (2)**
Singer-Songwriter	**14.3% (2)**
Vlaamse slagers	**14.3% (2)**
Alternative	**7.1% (1)**
Electronic	**7.1% (1)**
R&B	**7.1% (1)**
Rap	**7.1% (1)**

The future

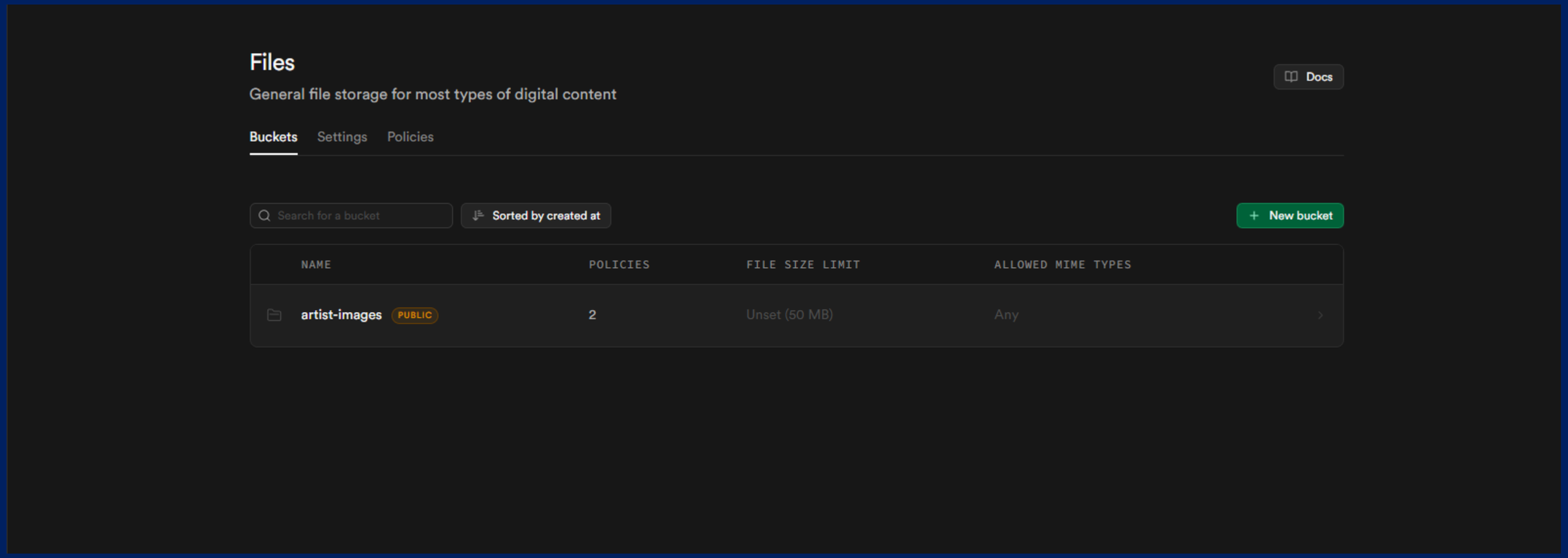
- Making it more accessible for more festivals
- Making it easier for the organisers to edit
- Upscaling

Appendix

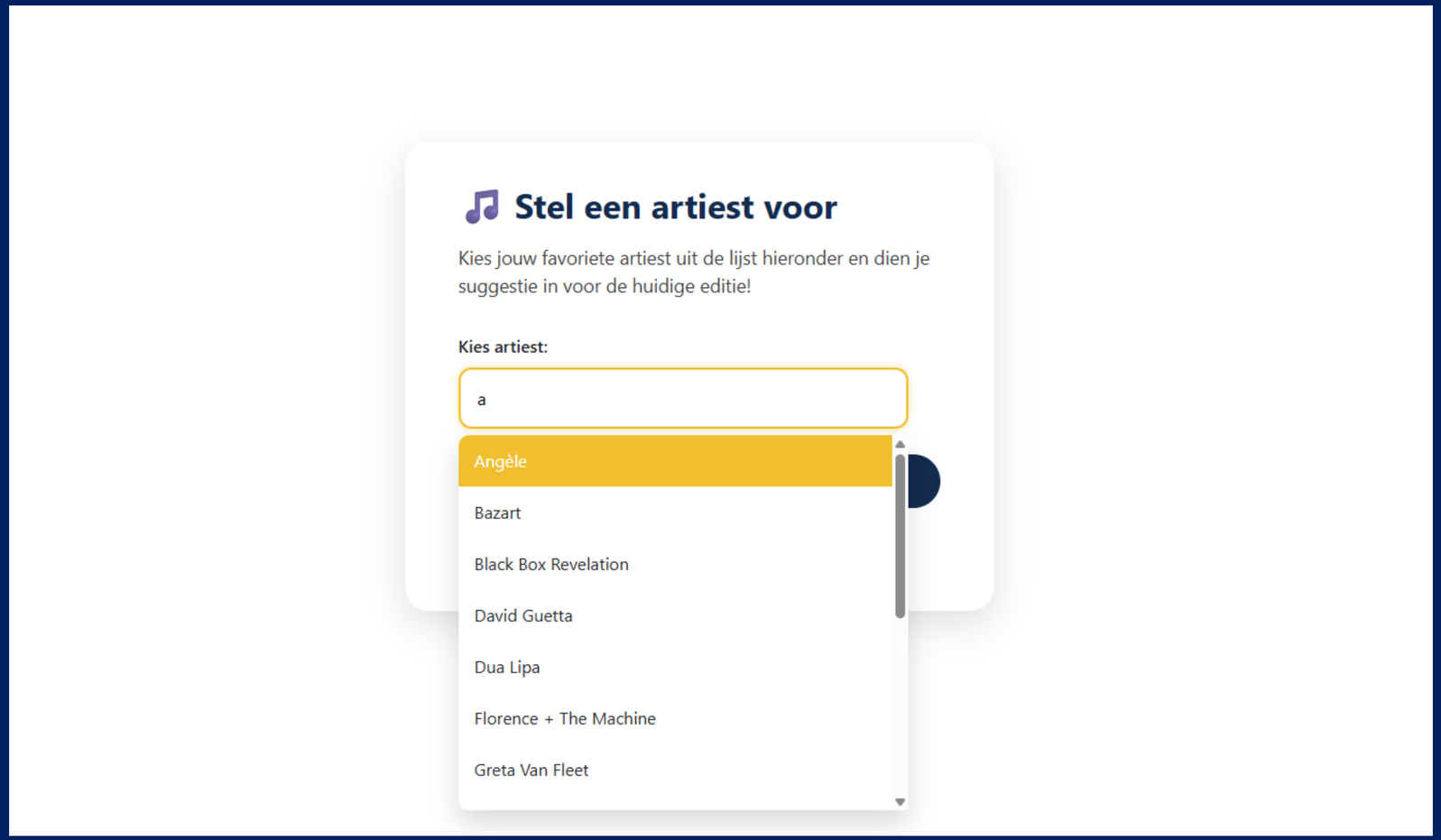
- Site Maspoe
- MVP render – polls
- Team 38 kanban - Miro (not possible to edit)


Genres
id bigint NN
name varchar NN
related_genre_id bigint
ArtistGenres
artist_id bigint NN
genre_id bigint NN
ArtistGenres_genre_id_fkey
ted_genre_id_fkey
ArtistGenres_artist_id_fkey
poll
id int NN
created_at timestamptz
Question text
festival_id int
is_visible bool NN
show_results bool NN
Polloption
id int NN
created_at timestamptz
text text
Count int
artist_id int
poll_id int
Artists
id bigint NN
created_at timestamp NN
Artist_name varchar NN
image_url varchar
edition_id int NN
Polloption_poll_id_fkey
poll_festival_id_fkey
Polloption_artist_id_fkey
Artists_edition_id_fkey
Suggestion_feedback_artist_id_fkey
FestivalEdition
id int NN
created_at timestamptz
Start_date date
End_date date
Name text
Location text
is_active bool NN
Suggestion_feedback
id int NN
artist_id int
user_id int
festival_id int
created_at timestamptz
is_hidden bool NN
Suggestion_feedback_festival_id_fkey
Votes_for_polloption_id_fkey
User
id int NN
created_at timestamptz
email text
is_admin bool NN
Suggestion_feedback_user_id_fkey
Votes_for
created_at timestamptz
user_id int NN
polloption_id int NN
Votes_for_user_id_fkey

Bucket for images



List of artist in supabase



Algorithmic Support

app > routes > poll.py > ...

```
137
138  @bp.get("/results")
139  def results():
140      user = get_session_user()
141      poll = get_or_create_active_poll()
142
143      # --- Muziekprofiel op basis van jouw suggesties (actieve editie) ---
144      genre_counts = {}
145      genre_percentages = {}
146
147      if user:
148          rows = (
149              db.session.query(Genres.name, func.count())
150              .join(ArtistGenres, ArtistGenres.genre_id == Genres.id)
151              .join(Artists, Artists.id == ArtistGenres.artist_id)
152              .join(SuggestionFeedback, SuggestionFeedback.artist_id == Artists.id)
153              .filter(SuggestionFeedback.user_id == user.id)
154              .filter(
155                  SuggestionFeedback.user_id == user.id,
156                  SuggestionFeedback.festival_id == poll.festival_id,
157              )
158              .group_by(Genres.name)
159              .all()
160          )
161          genre_counts = {g: c for g, c in rows}
162          total = sum(genre_counts.values()) or 1
163
164          genre_percentages = {
165              g: round(c * 100 / total, 1)
```

Algorithmic Support

app > routes > poll.py > results

```
139   def results():
165           g: round(c * 100 / total, 1)
166           for g, c in genre_counts.items()
167       }
168
169       # 🔥 Sorteer percentages aflopend
170       sorted_genres = sorted(
171           genre_percentages.items(),
172           key=lambda x: x[1],
173           reverse=True
174       )
175
176   # --- Jouw stemmen binnen de actieve editie ---
177   user_votes = []
178   if user and poll:
179       user_votes = (
180           db.session.query(VotesFor, Polloption, Poll, FestivalEdition, Artists)
181           .join(Polloption, VotesFor.polloption_id == Polloption.id)
182           .join(Poll, Polloption.poll_id == Poll.id)
183           .join(FestivalEdition, Poll.festival_id == FestivalEdition.id)
184           .join(Artists, Polloption.artist_id == Artists.id)
185           .filter(
186               VotesFor.user_id == user.id,
187               Poll.festival_id == poll.festival_id,
188           )
189           .order_by(FestivalEdition.Start_date.desc())
190           .all()
191       )
192
```

Algorithmic Support

app > services > genre_profile.py > generate_poll_for_user

```python
def generate_poll_for_user(user_id: int, num_options: int = 5):
    """Generate a personalized set of poll options for a user."""
    profile = get_user_genre_profile(user_id)
    proximity_scores = genre_proximity_scores(profile.keys())
    genre_id_lookup = {
        name: gid for gid, name in Genres.query.with_entities(Genres.id, Genres.name).all()
    }

    rows = (
        db.session.query(
            Artists.id,
            Artists.Artist_name,
            Genres.name.label("genre"),
        )
        .join(ArtistGenres, ArtistGenres.artist_id == Artists.id)
        .join(Genres, Genres.id == ArtistGenres.genre_id)
        .all()
    )

    artist_data = {}
    for artist_id, artist_name, genre in rows:
        if artist_id not in artist_data:
            artist_data[artist_id] = {
                "artist": artist_name,
                "genres": [],
            }
        artist_data[artist_id]["genres"].append(genre)
```

Algorithmic Support

app > services > genre_profile.py > generate_poll_for_user

```
29  def generate_poll_for_user(user_id: int, num_options: int = 5):
56
57      scored = []
58      for artist_id, data in artist_data.items():
59          genre_score = sum(profile.get(g, 0) for g in data["genres"])
60
61          # Calculate how close this artist's genres are to the user's favourites.
62          proximity = max(
63              (
64                  proximity_scores.get(genre_id_lookup.get(genre_name, -1), 0.0)
65                  for genre_name in data["genres"]
66              ),
67              default=0.0,
68          )
69
70          # Did user suggest it?
71          self_suggested = (
72              SuggestionFeedback.query
73              .filter_by(user_id=user_id, artist_id=artist_id)
74              .first()
75              is not None
76          )
77
78          score = 3 * int(self_suggested) + 2 * genre_score + 5 * proximity
79
80          scored.append((score, artist_id, data["artist"]))
81
82      scored.sort(reverse=True)
```